IndianOil

# INDIAN OIL CORPORATION LIMITED
## (Pipelines Division, NOIDA)

| INTER OFFICE MEMO | | | |
|---|---|---|---|
| From | DGM I/c (MNT), PLHO, Noida | To | DGM (Maint) WRPL Gauridad |
| Ref | PLHO/MNT/10.0 | Ref | |
| Date | 15.9.2015 | Date | |

**Sub: Investigation report on Breakdown of MP-2 at SMPL Sidhpur**

Subject investigation report is attached herewith; report consists of Probable Reasons of Breakdown, maintenance and operation issues, recommendations, Suggestions for Improvement. Suggestions of the committee are as under

- Hardness profile mapping of crank pins of the crankshaft having undersized crankpins.
- Periodic overhauling/refurbishment/replacement of engine critical components as per recommendations of OEM.
- Maintenance and replacement of radiator replacements.
- Avoidance of running of engines in stressed conditions for continuous periods.
- Re-rating of pump and engines.
- Availability of reports of major schedule and breakdown reports at Station, Base & regional office.

This is for your information and further needful action, please.

**(P.K.Bera)**

**Encl**: As mentioned above

Cc: GM (T) WRPL, GM ERPL
DGM WRPL Vadinar/Viramgam/Sendra/Chaksu/Mundra, PHBPL Haldia, NRPL Rewari
COM WRPL Vadinar/Jamnagar/Rajkot/Surendranagar/Viramgam/Sidhpur/Aburoad
/Kot/Sendra/Beawar/Rajola/Ramsar/Sanagner, PHBPL Bolpur/Haldia

# COMMITTEE REPORT ON THE FAILURE OF MP2 AT SMPL, SIDHPUR.

PLHO/MNT/Mech/1.0

Dated 11.8.2015

A committee of following officers was formed to investigate the breakdown of MP2 at SMPL, Sidhpur vide note dated 15.7.2015

1. Sh. C.S.Dutta, CPJM, PJ-Mech , PLHO, Noida
2. Sh. Harjeet Singh SPJM (T&I), PLHO, Noida
3. Sh. Sanjeev K. Jaiswal, MNM, PLHO, Noida

Scope of work:

- Root cause analysis of the incident
- Review of the maintenance practices both for engine and the protection system
- To identify the procedural lapse if any (both in operation & maintenance of the engine) specially in co-relation to the major engine failure of engine at Sidhpur in 2007 when severe crank shaft damage occurred & many recommendations were suggested for implementation

Committee has submitted its report vide note dated 31.7.2015 after duly carrying out the root cause analysis of the incident, review of the existing maintenance and operation procedures and practices.

As brought out by the committee, the possible cause of failure of the engine are as follows:

1. Prolonged running of the engine with high exhaust temperature (At 476 degree against maximum recommended temperature of 430 Degree C) and at high lube oil temperature ( 75 degree C against maximum recommended temperature of 70 degree C)
2. Choking of elements in the radiator (charge air, water & lube oil) resulted ineffective heat transfer of the elements
3. Engine components like connecting rods for few cylinders have run more than 60000 hrs. against recommended refurbishment of connecting rods at an interval of 27000 hrs. Indicating lapse in maintenance practices.
4. Hardness profile mapping has not been undertaken since last major breakdown maintenance in 2007. In 2007, the hardness was in the range of 350. In the intervening period the hardness of the crank pins might have increased beyond permissible limit of 350 BHN ( Hardness observed after breakdown is in the range of 531 to 633 at crank pin no. 1,2,5 & 8)

Following recommendations have been proposed by Committee in view of the said incident of engine breakdown:

1. **Hardness profile mapping** of engine crankpins are to be carried out at regular interval (preferably in major maintenances in which connecting rods are removed) for **all engines having undersized crankpin**.
2. **Periodic overhauling/ refurbishment/ replacement of engine critical components** are to be adhered as per OEM guidelines. Special and due care to be taken for engine components, which are under cyclical load such as connecting rod and its components.
3. Considering the importance of cooling of lube oil, charge air and water in any engine, due care to be taken to **replace** these **radiator elements** at regular intervals.

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4. **Engine should not run in stressed condition** for continuous periods, all parameters of engines are to be adhered and strict monitoring is to be carried out for any deviation of engine parameters.
5. Necessary considerations for **re-rating of pump and engine** due to aging factor may be looked into for trouble free/ reliable operation.
6. All breakdown and schedule maintenance reports should be available at station, base and region.
7. Considering huge number of engines being operated in Western Region, system is to be developed to keep documentary evidence of all defective engines, which has past record of failure.

The committee recommendation is hereby recommended for approval. On approval Regions will be advised for implementation of the recommendation for improvement in the operation & maintenance practice.

11/8/2015

T.K.Bera
DGM I/C (Maintenance)

**~~ED (O)~~** 14/8/15

In addition to ER & WR offices.

DGM I/C (M&I): Please circulate to all SICs, where ever engine is the prime mover. Some of the observations in general way will be applicable to DG sets & F.F. engines.

14/8/15

31.7.2015

# DETAILED REPORT ON THE FAILURE OF MP-2 OF SMPL, SIDHPUR

## 1. Preamble:

This is in reference to the formation and approval of committee for investigation of failure of MP-2 at SMPL Sidhpur. Scope of work for the committee is as under

i. Root cause analysis of the incident.

ii. Review of the maintenance practices both for engine and the protection system.

iii. To identify the procedural lapse, if any, (both in operation and maintenance of the engine) specially in co-relation to the major engine failure of engine at Sidhpur in 2007 when severe crank shaft damage occurred & many recommendations were suggested for implementation.

## 2. Specification of MP-2:

| | | |
|---|---|---|
| Make | : | Allen |
| Engine serial no. | : | 8S37G/D-6/50161-1-1/1-1 |
| Engine type | : | 8S37G |
| Running hrs | : | 86707 |
| BHP | : | 2951** |
| Bore * Stroke | : | 320 * 370 mm |
| Fuel | : | LS Crude Oil |
| Speed range | : | 375 to 750 rpm |

** MP-2 pumping unit at Sidhpur was installed and commissioned on 14.07.2004. This engine was shifted to SMPL, Sidhpur from KBPL, Kandla. Radiators and Mainline pump of old MP-02 were used. Secondary water-cooling systems of the engine were removed and direct cooling as present in the old engine was retained after certain modification. Brief history of various maintenances of MP-2 carried out at Sidhpur are given below:

## 3. Maintenance history of the engine:

The following major schedule maintenances have been carried out in this engine till date at Sidhpur:

| S. No. | Type of Major Maintenance | Running Hours |
|---|---|---|
| 1 | 12000 hrs | 32685 |
| 2 | 6000 hrs | 39223 |
| 3 | 24000 hrs | 45081 |
| 4 | 6000 hrs | 51787 |
| 5 | 12000 hrs | 58137 |

CDutta 31/7/15
(C. S. Dutta)
CPJM (Mech)

31/7/15
Harjeet Singh

Sanjeev

| S. No. | Type of Major Maintenance | Running Hours |
|---|---|---|
| 6 | 6000 hrs | 65491 |
| 7 | 24000 hrs | 72526 |
| 8 | 6000 hrs | 78188 |
| 9 | 12000 hrs | 86707 |

4. **Chronology of events for present breakdown of engine:**

a. MP-2 was not available for operation since 1020 hrs on 19.6.2015.

b. Engine was not available due to DE side leakage from seal cooling pipe.

c. Rectification work was carried out and engine was started on 1452 hrs 24.6.2015.

d. The engine had tripped in 100% OMD deviation in cylinder 3 at 2307 hrs on 24.6.2015.

Engine parameters, such as lube oil, charge air, jacket water pressure and temperature found normal up to 2200 hrs as noted on operation log book.

**Observations after breakdown are as under:**

i. After MP-2 tripped in 100% OMD in cylinder 3, first the temperature of big and bearing and main bearing of all cylinders were checked from crank case doors by temperature gun

ii. All crank case doors were opened.

iii. Temperature of connecting rod 2 at big end bearing position was found 95 deg C. However, other bearing temperatures were found in the range of 78-82 deg C.

iv. All main bearing temperatures were checked and found to be in the range of 75-80 deg C.

v. White metal was found stuck on crank pin no. 2. Severe damage was noticed in both the halves of Big end bearing. Heat marks noticed on outer and inner side of bearing and connecting rod.

vi. Free spread of bearing no. 2 was negative

vii. Connecting rod no. 2 also have heat marks

viii. Damage in big end bearing no.8 observed.

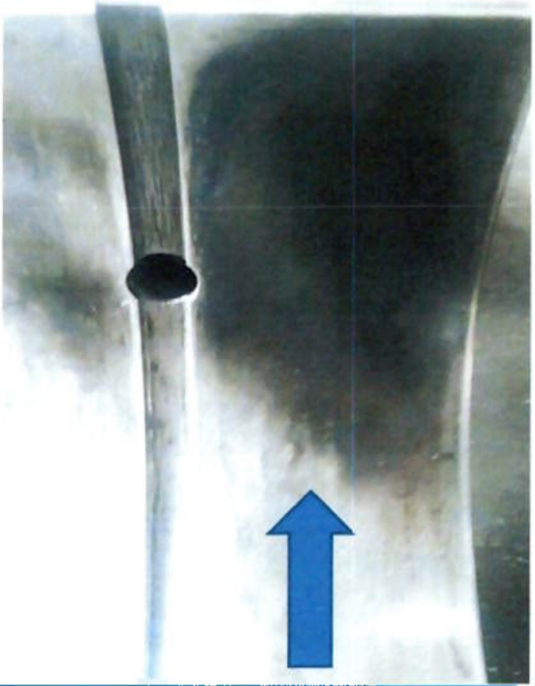
ix. M/s Exacto Technology, Coimbatore has carried out hardness profile mapping and Magnetic Particle Inspection (MPI) test on all crankpin. Hairline cracks have been observed on crankpin no. 1, 5 & 8 during MPI.

x. Hardness of 1, 2, 5 & 8 crankpin was found more than 350 HB; in some area the hardness found to the tune of 600 HB.

CSDutta
(C. S. Dutta)

31/7/15

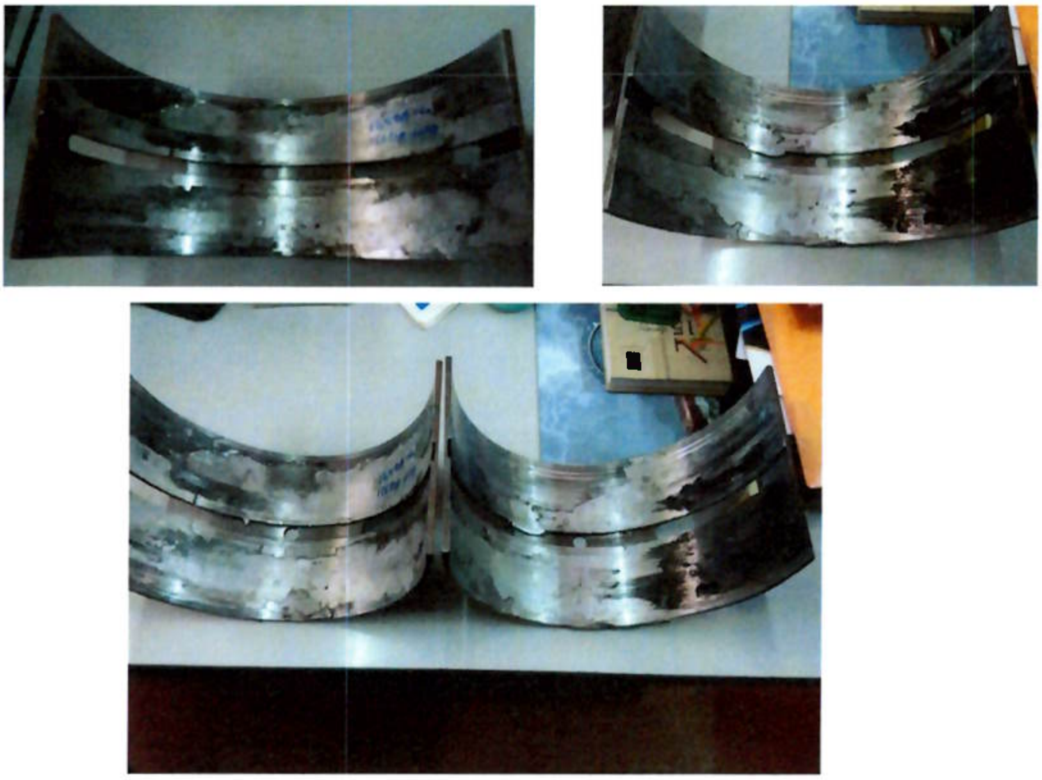
Sanjeev



Heat marks on connecting rod

Babbitt metal stuck on crank pin

**Scoring and heat marks on Connecting Rod & Crank Pin ( Cy.No.2)**



**Big End Bearing No.2**

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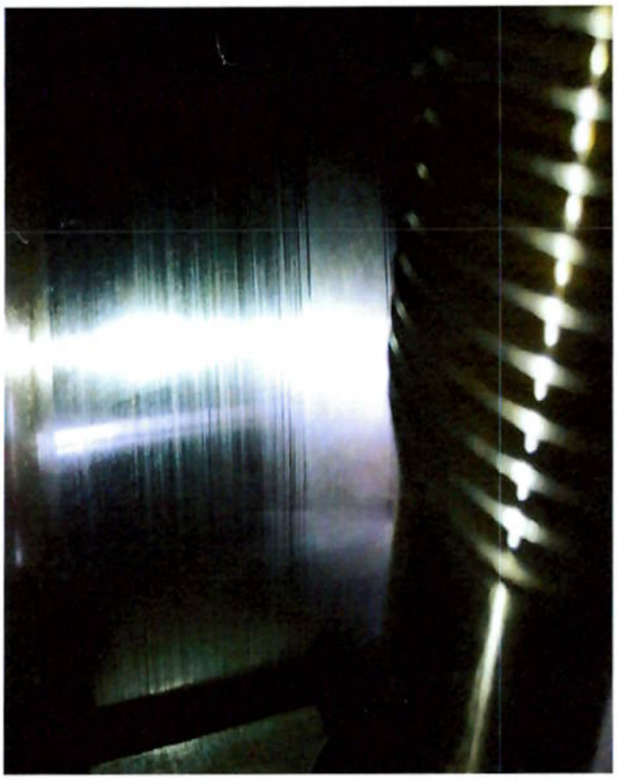
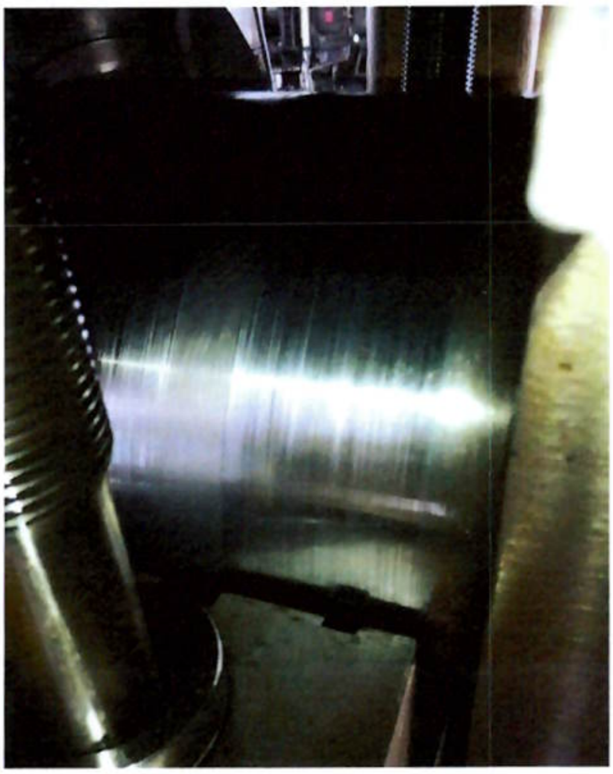
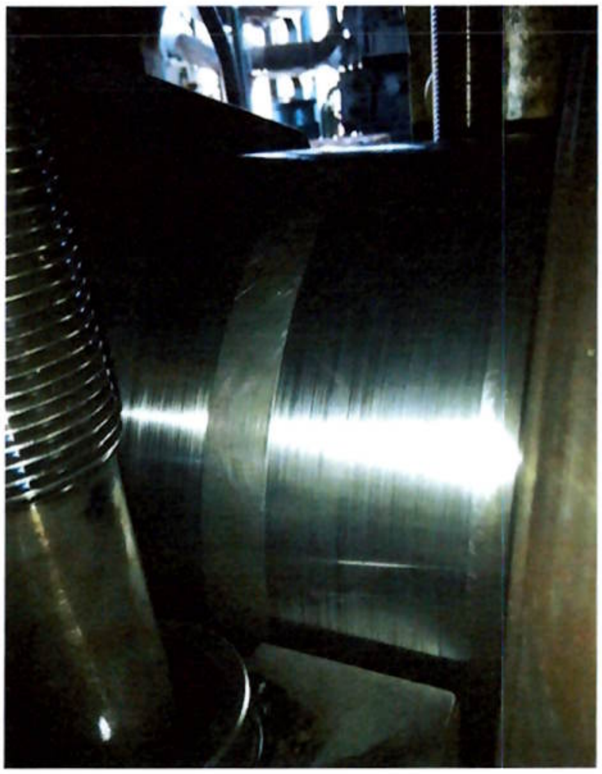
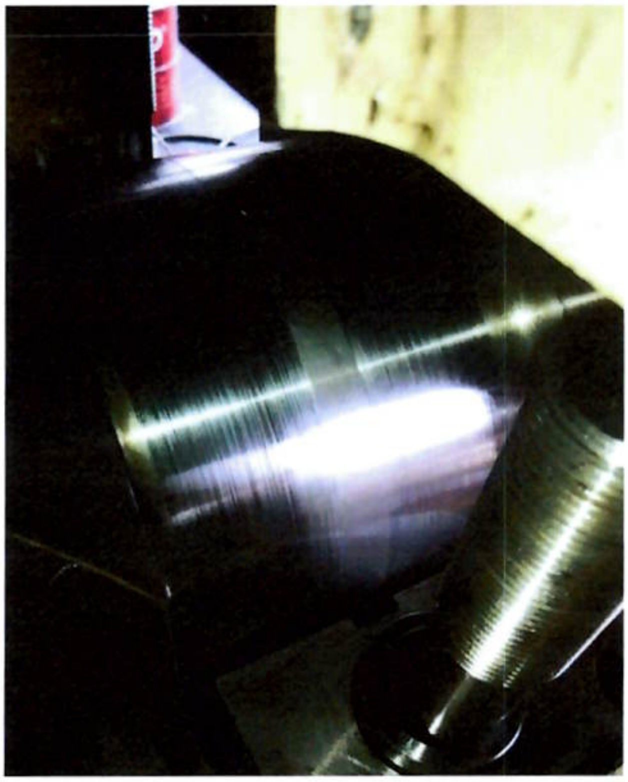




Crank Pin Cylinder 1 & 5



Crank Pin No.8



Scoring marks on Journal

31/7/15

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## 5. Observation of present committee:

Committee has visited site on 20.7.2015 & 21.7.2015 and reviewed following reports:

1. MP-2 Refurbishment report - July 2004 (Kandla engine shifted to Sidhpur).
2. Breakdown maintenance report - April 2007.
3. Last 12000 hours major maintenance report - June 2014.
4. Operation log books, defect register, lube oil and water consumption registers.
5. M/s EXACTO Technology, Coimbatore report on MPI and hardness profile mapping
6. Committee report dated 02.7.15 on present breakdown maintenance

### 5.1. Last Major Breakdown Maintenance of 2007:

#### 5.1.1. Reason for breakdown

Engine could not be stopped on low suction, though stop command was given from control room. Engine was tried to be stopped by switching off AMOT panel. Engine speed jumped up to 955.4 RPM and finally engine was stopped by cutting off fuel supply. Committee report attached as **Annexure A** (Unsigned copy of report received from WRPL Sidhpur vide e-mail dated 16.7.2015, Signed Report not available at site.)

#### 5.1.2. Observations of Rolls Royce on 2007 breakdown:

i. The crank pin of unit no 1, 5 and 8 has suffered severe rubbing marks due to the movement of bearing caused due to over speed of the engine.

ii. The crankpin of unit no 1 and 8 had surfaced chipping. The depth of chipping will be checked after polishing of the crank pin. The IOCL was requested to send the dimension and hardness readings after polishing the crank pin.

iii. Rough measurement of depth of chipping was recorded by IOCL supplied depth gauge. The depth of crank pin no 8 chipped area was observed roughly around 0.35 mm, and for unit no 5 it was 0.15 mm. The depth of chipped area of crank pin no 1 was difficult due its position.

#### 5.1.3. Works carried out by M/s Eurotech Systems, Coimbatore

Subsequent to above incident, M/s Eurotech Systems, Coimbatore was engaged to do repair work on crank pins and carry out MPI on crankshaft.

i. M/s Eurotech pin at different points. Ovality was observed in all the three crankpin (Reports attached).

ii. Run out of main journal number 6 and 7 also checked and found that it was above limit. Run out reading taken by IOCL was also provided to Eurotech systems.

iii. Initially surface cutting of crankpin No 1, 5 and 8 carried out to partially release the surface stress.

Sd/- 31/7/15 Sanjeev

iv. After that straightness of all main journal were checked and reading of all the main journal was above limit (Report attached as Annexure I).

v. Peening in crankpin No 1, 5 and 8 carried out to bring down the run out reading. After peening the straightness is again checked as per Eurotech Systems procedure and run out reading of crankshaft is attached as Annexure-I.

vi. Main journal No 10 have maximum run out reading of 0.085 mm after peening. Therefore oil clearance of that journal is checked and it is found in between 0.21 mm to 0.225 mm i.e. well within acceptable limit.

vii. After making the crankshaft straight, grinding of crankpin No 1, 5 and 8 carried out to remove the peening marks and rubbing marks.

viii. After grinding, radius grinding, oil hole trimming and polishing carried in crankpin No 1, 5 and 8.

ix. After grinding following parameters have been checked:

x. Crankpin dimensions are measured and the Ovality and taper of ground pin No 1, 5 and 8 is maintained within 0.015 mm (Reports attached).

xi. No cracks were detected during MPI and DP inspection.

### 5.2. Recommendations of the committee (Breakdown report of 2007) and compliance

| Recommendations | Observations Made during site visit |
|---|---|
| Calibration / checking of all the governors at the time of major maintenance like 18000 hrs (for New G-series engines) and 12000 hrs (for Old G-series engines). | Maintenance crew confirmed to have checked governor during schedule maintenance. However, records are not available for verification. |
| Checking of proper working of the butterfly valve for complete opening and closure at the time of major maintenance like 18000 hrs (for New G-series engines) and 12000 hrs (for Old G-series engines). | Maintenance crew confirmed to have checked butterfly valve for complete opening/closing during schedule maintenance in June 2014 (12000 hrs). However, records are not available for verification. |
| Checking the proper working of the start/ stop command logics by simulating start/ stop actions at regular pre-defined intervals. | Checking is done as per IMS format SF/TII/19 (Format for schedule maintenance of instrumentation system of Mainline Engine/Pump (G Series) |
| Checking of the operating time of the suction and discharge valves at the time of major maintenance like 18000 hrs (for New G-series engines) and 12000 hrs (for Old G-series engines). This is required because due to continuous use the travel time of the valve may increase, which may have an adverse impact on the operations. | Maintenance crew confirmed to have checked. However, records are not available for verification. |
| Stop logic may be changed; suction and discharge gate valves (MOV's) should get a stop command only when the engine speed reaches around 200 rpm as per the manufacturer's manual. | Implemented. |

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## 6. Observations of the present Committee:

### 6.1. Operating parameters of Engine:

i. Exhaust temperatures of cylinder heads should be less than 430 deg C. However, the same were found higher than the limit as recorded in the log book. As per log book data on 12.6.2015, it has gone up to 476 deg C in cylinder 3, on 13.6.2015 it has gone up to 464 deg C in cylinder 3, on 14.6.2015 it has gone up to 470 deg C in cylinder 3. Engine has run consistently with high exhaust temperature.

ii. It is observed that charge air temperature was often found to be in the range of 70-75 deg C whereas the normal temperature should be below 66 Deg C.

iii. Lube oil temperature was also found around 75 Deg C. To counter this, heat exchanger has been added in the lube oil system in addition to the existing cooling through radiator.

### 6.2. Maintenance Issues:

i. Above operation parameters indicates choking of elements resulting in in-effective heat transfer of these elements. External conditions of elements are in poor condition, having scaling and rusting marks on it. These elements are also repaired several times.

ii. Engine components such as connecting rods for some cylinders have run for more than 60000 hrs. against recommended refurbishment of connecting rods at an interval of 27000 hrs.

Due to above stated reasons water is continuously sprayed over the elements to maintain the operating parameters.

## 7. Probable cause of failure

After reviewing the documents such as the Breakdown maintenance report of 2007, Committee report dated 2.7.2015, operation log book, defect register, maintenance history of MP-2 and after interacting with the shift & maintenance personnel, following probable causes might have contributed to breakdown of engine:

### 7.1. Issues related to last major breakdown maintenance of 2007

a. During last major breakdown maintenance of 2007, under sizing of crankpin 1, 5 & 8 was done up to 1.0 mm, but some very light rubbing marks were still there in all the three crankpins. It was suggested by M/s Eurotech to go for grinding up to 1.5 mm to ensure complete removal of rubbing marks on these crank pins.

b. During last major breakdown maintenance of 2007, hardness of the crankpin was checked by M/s Eurotech Systems, Coimbatore after the grinding and it was found less than 320 HB (Limits New Shaft – 220, Old/Rebuilt Shaft – 350).

c. During present breakdown, hardness profile mapping has been carried out by M/s Exacto Technology, Coimbatore in which hardness profile noted is as under:

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| Crankpin no. | 1 | 2 | 5 | 8 |
|---|---|---|---|---|
| Max. Hardness in BHN | 531 | 605 | 633 | 598 |
| Min. Hardness in BHN | 202 | 202 | 200 | 195 |

d. Hardness profile mapping is not undertaken since last major breakdown maintenance of 2007. In the intervening period, the hardness of the crankpins might have increased beyond permissible limit of 350 BHN.

7.2. **Maintenance issue:**

a. Connecting rods of MP-2 engines have crossed **recommended life** of refurbishment @ 27000 hrs. (Cumulative Running Hours of con rod no.2 is >60000 hrs.) , which might have resulted in twisting of connecting rod no.2 (also mentioned in committee report of 2.7.15). This might have resulted in generation of excessive stress on connecting rod, crank pin and subsequent failure.

b. Choking of radiator elements (charge air, water and lube oil) resulting in in-effective heat transfer through these elements. External conditions of elements are also in poor condition, having scaling and rusting marks on it. These elements are also repaired several times.

7.3. **Operating condition:**

a. High cylinder exhaust temperature (above $430^0$ C) and charge air temperature (above $66^0$ C) might have resulted in stressed running of engine resulting in excess loading of engine components such as connecting rod, big end bearing and crankpin. This might have resulted in deterioration of engine components, which are under cyclical load condition.

b. Engines are running @ 95-100% or even higher loading condition in most of the cases. This might have resulted in failure of critical engine components.

8. <u>RECOMMENDATIONS:</u>

i. **Hardness profile mapping** of engine crankpins are to be carried out at regular interval (preferably in major maintenances in which connecting rods are removed) for **all engines having undersized crankpin**.

ii. **Periodic overhauling/ refurbishment/ replacement of engine critical components** are to be adhered as per OEM guidelines. Special and due care to be taken for engine components, which are under cyclical load such as connecting rod and its components.

iii. Considering the importance of cooling of lube oil, charge air and water in any engine, due care to be taken to **replace** these **radiator elements** at regular intervals.

31/7/15

Sanjeev

iv. **Engine should not run in stressed condition** for continuous periods, all parameters of engines are to be adhered and strict monitoring is to be carried out for any deviation of engine parameters.

v. Necessary considerations for **re-rating of pump and engine** due to aging factor may be looked into for trouble free/ reliable operation.

vi. All breakdown and schedule maintenance reports should be available at station, base and region.

vii. Considering huge number of engines being operated in Western Region, system is to be developed to keep documentary evidence of all defective engines, which has past record of failure.

31/7/15 31/7/15 Sanjeev Kumar

| | | |
|---|---|---|
| **(Chandra Sekhar Dutta)** | **(Harjeet Singh)** | **(Sanjeev Jaiswal)** |
| **CPJM (Mech)** | **SPJM(T&I)** | **MNM** |

**DGM ~~I/C~~ (Maint)** Separate note attached.

Sanjiv Kumar

**JAISWAL, SANJEEV (संजीव जायसवाल)**

| | |
|---|---|
| **From:** | Das, Parichay (दास,परिचय) |
| **Sent:** | Thursday, July 16, 2015 4:50 PM |
| **To:** | JAISWAL, SANJEEV (संजीव जायसवाल) |
| **Cc:** | Kumar, Ashok (कुमार, अशोक) |
| **Subject:** | FW: MP-2 SMPL Sidhpur |
| **Attachments:** | runout reading.jpg; straightness check.xls; 1.bmp; 2.bmp; Chronology of grinding activities MP-2.doc; Crankpin grinding Sidhpur note-Draft Note from Sidhpur.doc; cy head.xls; DETAILED REPORT ON THE FAILURE OF MP-2 SMPL SIDHPUR1.doc; IOCL LETTER.bmp; letter.doc; Minutes of Meeting held between Rolls Royce.doc; Minutes of Meeting held betweenEurotech Systems.doc; MP02 insitu grinding.ppt; MP-02 restoration work.ppt; MP-2 CRANK SHAFT RUN OUT DATA -PEENING.doc; MP-2 Restoration schedule as on 30.5.07.doc; RUN OUT READING.doc; runout checking.jpg; runout reading 1.jpg |

Respected Sir,

This is in line with the telephonic conversation had with you just now. We are sending you all related details related to breakdown maintenance of MP-2 (peening and in-situ CR pin grinding) which was carried out in 2007.

Submitted for your kind information please.

Yours sincerely,
Parichay Das
Senior Engineer (O&M)
WRPL, Sidhpur
Mo: +91-9428827993

**From:** Das, Parichay (दास,परिचय)
**Sent:** 16 July 2015 16:35
**To:** JAISWAL, SANJEEV (संजीव जायसवाल)
**Cc:** Kumar, Ashok (कुमार, अशोक)
**Subject:** RE: MP-2 SMPL Sidhpur

Respected Sir,

Report on 12000 Hrs scheduled maintenance of MP-2 Sidhpur carried out from 09.06.2014 to 10.06.2014 is enclosed herewith. All IMS formats pertaining to the maintenance is also enclosed for your kind perusal.

Submitted for further review and your wistful accordance please.

Yours sincerely,
Parichay Das
Senior Engineer (O&M)
WRPL, Sidhpur
Mo: +91-9428827993

**From:** JAISWAL, SANJEEV (संजीव जायसवाल)
**Sent:** 16 July 2015 16:23
**To:** Das, Parichay (दास,परिचय)
**Cc:** Kumar, Ashok (कुमार, अशोक); DUTTA, CHANDRA SEKHAR (चन्द्र शेखर दत्ता); SINGH, HARJEET (हरजीत सिंह)
**Subject:** MP-2 SMPL Sidhpur

Please forward previous major maintenance report for subject MLPU urgently.

Warm Regards

Sanjeev K Jaiswal
Maintenance Manager
Pipeline Head Office, Noida
Tel: 0120-2448904
Fax: 0120-2448024

## DETAILED REPORT ON THE FAILURE OF MP-2 OF SMPL, SIDHPUR

### INTRODUCTION:

MP-2 pumping unit at Sidhpur was installed and commissioned on 14.07.2004. This engine was shifted to SMPL, Sidhpur from KBPL, Kandla. Radiators and Mainline pump of old MP-02 were used. Secondary water-cooling systems of the engine were removed and direct cooling as present in the old engine was retained after certain modification. Brief history of various maintenances of MP-2 carried out at Sidhpur is given below:

### MAINTENANCE HISTORY OF THE ENGINE:

The following major schedule maintenances have been carried out in this engine till date at Sidhpur :

| S. No. | Type of Major Maintenance | Running Hours |
|---|---|---|
| 1 | 12000 hrs | 32685 |
| 2 | 6000 hrs | 39223 |
| 3 | 24000 hrs | 45081 |

24000 hrs of MP-2 was recently completed on 24.03.2007 (No load test was completed on 17.03.2007 & the load test was carried out on 24.03.2007). Since then the engine has run about 62 hrs.

On 11.04.2007 engines tripped at Viramgam, subsequently, Sidhpur faced a problem of low suction. Following is the sequence of events :

| S.No. | TIME | DESCRIPTION OF THE EVENT | REMARKS |
|---|---|---|---|
| (1) | 18.40 hrs | MP-1 of Viramgam station tripped subsequently MP-3 was also stopped at 18.47 hrs | |
| (2) | 18.45 hrs | Speed of MP-2 & MP-3 (Sidhpur) was reduced on auto. | |
| (3) | 18.49.16 hrs | Mainline inlet suction pressure = 2.5 $Kg/cm^2$<br>Suction header pressure (PS 44) = 2.3 $Kg/cm^2$ | |
| (4) | 18.49.17 hrs | MP-2 stop command was given from the control room. | MP-2 did not stop, however, by the subsequent events it can be construed that the suction & discharge MOV's have started closing. |

| S.No. | TIME | DESCRIPTION OF THE EVENT | REMARKS |
|---|---|---|---|
| (5) | 18.49.18 hrs | Low suction alarm was annunciated from PS-44 (suction header) | Low suction alarm setting is 2.0 $Kg/cm^2$ |
| (6) | 18.49.20 hrs | Low suction alarm was annunciated from PT-45 (suction header) | |
| (7) | 18.49.25 hrs | Engine suction pressure low alarm was annunciated from PS- 16 A (suction arm of MP-2) through Amot Panel.<br>Engine second stage fault MP-2 appeared. | Low suction alarm setting is 1.5 $Kg/cm^2$<br><br>Execution of stop command, confirmed logics working up to 3 way pneumatic valve for common second stage faults in MP-2 Amot panel (VCG-48) etc., |
| (8) | 18.49.26 hrs | Mainline inlet suction pressure came down to 1.2 $Kg/cm^2$ & the suction header pressure to 0.9 $Kg/cm^2$ | MP-2 continued to run & the speed started increasing from 532.9 to 542.2 rpm. |
| (9) | 18.49.27 hrs | MP-2 in fault alarm appeared | Amot panel was switched off. |
| (10) | 18.49.36 hrs | Mainline inlet suction pressure came down to 0.5 $Kg/cm^2$ & the suction header pressure was 0.4 $Kg/cm^2$ | MP-2 speed increased to 543.6 rpm. and no flow condition was reached resulting in a No Load type condition for the engine |
| (11) | 18.49.46 hrs | MP-2 in local UCP, engine first stage fault MP-2, MP-2 common 1st stage fault appeared. Suction pressure dropped further to 0.2 $Kg/cm^2$ at both the mainline inlet and the suction header. | MP-2 speed increased to 547.9 rpm and no flow/ no load type condition of the engine persisted. |
| (12) | 18.49.56 hrs | The suction pressure at the mainline inlet & suction header started showing an increasing pattern | This indicates that the suction & discharge valves of MP-2 were in the process of closing. |
| (13) | 18.50.16 hrs | Mainline inlet suction pressure further increased to 4.8 $Kg/cm^2$ & the suction header pressure was 4.6 $Kg/cm^2$ | MP-2 speed increased to 655.5 rpm |
| (14) | 18.50.21 hrs | Engine speed HI & Engine speed HI – HI for MP-2 appeared simultaneously | |
| (15) | 18.50.26 hrs | Mainline inlet suction pressure was observed as 3.5 $Kg/cm^2$ & the suction header pressure was 3.3 $Kg/cm^2$ | MP-2 speed jumped to 955.4 rpm (There was huge increase of 384.7 rpm in MP-2 within 20 |

| S.No. | TIME | DESCRIPTION OF THE EVENT | REMARKS |
|---|---|---|---|
| | | | seconds from 18.50.06 hrs to 18.50.26 hrs) |
| (16) | 18.52.47 hrs | Mainline inlet suction pressure was observed as 15.8 Kg/cm$^2$ & the suction header pressure was 15.7 Kg/cm$^2$ | MP-2 was finally stopped by fuel cut-off from the fuel forwarding module by closing the valve. |

After reviewing all the documents such as the Operation log book, Defect Register, maintenance history of MP-2, PLC data etc., after interacting with the shift & maintenance personnel and the circumstantial evidence, following probable causes have been identified which might have contributed in non stopping of MP-2 :

(1) Non activation/ malfunctioning of the 3-way valve (VCG-31) in Amot Panel of MP-2, when the stop command was given from the control room {refer event no. (4)}, which cuts the air supply to the engine governor.

(2) Non-activation/ malfunctioning of the governor bellow of MP-2 when the stop command was given from the control room {refer event no. (4)}, which cuts the fuel supply to the engine.

(3) The butterfly valve has not closed completely after the switching off the Amot panel {refer event no. (9)} thereby continuing to supply a small amount of charge air to the engine, which may be adequate for the engine as by that time the engine was running in no flow/ no load type condition.

**OBERVATIONS** :

(1) Due to any one of the above cited reasons MP-2 did not stop and continued to run when the stop command was given from the control room, while the suction and discharge valves started closing.

(2) When Amot panel was switched off, the air supply gets cut to the Engine Governor. It also closed the butterfly valve, which cuts-off the charge air supply to the engine. Notwithstanding the above the engine did not stop {refer event no. (9)} as by that time no flow/ no-load type condition was achieved.

(4) Hi-speed and Hi-Hi speed alarm also failed to stop the engine as by that time the engine has achieved no flow/ no-load type condition.

(5) Mechanical over speed trip was found operated during physical inspection of engine after the incident.

(6) Governor shut down rod setting, which trips the engine upon loss of control air was found to be disturbed. As a result the engine did not stop even after switching off the control air supply from the Amot panel.

**ASSESSMENT OF THE EXTENT OF DAMAGE TO THE ENGINE :**

All the cylinders were checked and the observations are as under :

(1) Big end bearing of cylinder no. 1,5 & 8 was found damaged and these are beyond repair. Big end bearing of cylinder no. 2,3,4,6 & 7 are OK and the same has been fitted back.

(2) Connecting rod of cylinder no. 1,5 & 8 have heating marks. These three connecting rods and the piston assembly were dismantled and the small end bearing and the gudgeon pin were also checked but no abnormality was found in the small end bush and the gudgeon pin. These three connecting rods can be reused after repair.

Some white metal deposit was observed on the crank pin no. 1, 5 & 8. After cleaning the white metal from the surface of all the three crank pins have been removed.

Connecting rod of cylinder no.2,3,4,6 & 7 were checked and found OK.

(4) Pump casing was removed and no damage has been found on the impeller vanes, wear rings, throat bush etc., The pump rotor was removed and fitted back after cleaning.

The governor has to be overhauled and calibrated at Gauridad before putting MP-2 back to operation. Charge air butterfly valve shall be checked for normal operation.

Magnetic particle inspection of the affected connecting rods no.1, 5 & 8 has to be carried out through an external agency to ascertain the presence of surface cracks, if any. Further, the main bearing of these cylinders should also be checked.

MP-2 is expected to be ready for operations by 21.04.2007.

**RECOMMENDATIONS :**

Based on the above observations following actions are proposed to avoid occurrence of such incidences in future :

(1) Calibration/ checking of all the governors at the time of major maintenance like 18000 hrs (for New G-series engines) and 12000 hrs (for Old G-series engines).

(2) Checking of proper working of the butterfly valve for complete opening and closure at the time of major maintenance like 18000 hrs (for New G-series engines) and 12000 hrs (for Old G-series engines).

(3) Checking the proper working of the start/ stop command logics by simulating start/ stop actions at regular pre-defined intervals.

(4) Checking of the operating time of the suction and discharge valves at the time of major maintenance like 18000 hrs (for New G-series engines) and 12000 hrs (for Old G-series engines). This is required because due to continuous use the travel time of the valve may increase, which may have an adverse impact on the operations.

(5) Stop logic may be changed, suction and discharge gate valves (MOV's) should get a stop command only when the engine speed reaches around 200 rpm as per the manufacturer's manual.

**(RAJESH UPRETY)**
**SMNM, VIRAMGAM**

**(A.K.DOGRA)**
**SM (T&I), VIRAMGAM**

**(PROMOD CHELLI)**
**DMNM, GAURIDAD**

**(TOMY ZACHERIA)**
**ST&IE, SIDHPUR**

**(AMIT KUMAR)**
**SO&ME, SIDHPUR**

**Minutes of Meeting held between Rolls Royce, Delhi and Indian Oil Corp. at SMPL Sidhpur pump station on 27th June 2007**

**Subject: Insitu crankpin Grinding in MP-02**

| **For Rolls-Royce** | **For IOCL** |
|---|---|
| Mr. Amit Nagar | Mr. D. K. Banerjee<br>Mr. Amit Kumar |

1.Rolls-Royce representative arrived at pump station on 21st April, 07 to analyze the failure of engine MP-2.

2.The crank pin of unit no 1, 5 and 8 has suffered severe rubbing marks due to the movement of bearing caused due to over speed of the engine.

3. The crankpin of unit no 1 and 8 had surfaced chipping. The depth of chipping will be checked after polishing of the crank pin. The IOCL was requested to send the dimension and hardness readings after polishing the crank pin.

4. Rough measurement of depth of chipping was recorded by IOCL supplied depth gauge. The depth of crank pin no 8 chipped area was observed roughly around 0.35 mm , and for unit no 5 it was 0.15 mm. The depth of chipped area of crank pin no I was difficult due its position.

5. The hardness of crank pin of unit no5 will be known after polishing the crank pin.

6. The IOCL was advised to check the condition of linkages and fuel racks start limit position after installing the new governor.

7. The IOCL was advised to check the condition of push rods and exhaust inlet valves, push rods and cam lobe of all cylinders.

8. Rolls-Royce representative left site on 21st April.07

For Rolls- Royce

For IOCL

**Minutes of Meeting held between Eurotech Systems, Coimbatore and Indian Oil Corp. at SMPL Sidhpur pump station on 27th June 2007**

**Subject: Insitu crankpin Grinding in MP-02**

| **For Eurotech Systems** | **For IOCL** |
|---|---|
| Mr. Krishna Kumar<br>Chief Engineer | Mr. Amit Kumar<br>Senior Engineer |

1. Eurotech systems representative arrived at pump station on 14th May, 07 to carry out repair work of MP-02 by in-situ grinding and peening of crankpin of engine.

2. The crank pin of unit no 1, 5 and 8 has suffered severe rubbing marks due to the movement of bearing caused due to over speed of the engine.

3. Eurotech systems had carried out the MPI test, DP test, Hardness test and checked the dimension of the crank pin at different points. Ovality was observed in all the three crankpin (Reports attached).

4. Run out of main journal number 6 and 7 also checked and found that it was above limit. Run out reading taken by IOCL was also provided to Eurotech systems.

5. Initially surface cutting of crankpin No 1,5 and 8 carried out to partially release the surface stress.

6. After that straightness of all the main journal were checked and reading of all the main journal was above limit (Report attached as Annexure I).

7. Peening in crankpin No 1, 5 and 8 carried out to bring down the run out reading. After peening the straightness is again checked as per Eurotech Systems procedure and run out reading of crankshaft is attached as Annexure-I.

8. Main journal No 10 have maximum run out reading of 0.085 mm after peening. Therefore oil clearance of that journal is checked and it is found in between 0.21 mm to 0.225 mm ie well within acceptable limit.

9. After making the crankshaft straight, grinding of crankpin No 1, 5 and 8 carried out to remove the peening marks and rubbing marks.

10. All the peening marks were removed while making the crankpin 1.0 mm undersize but some very light rubbing marks are still there in all the three crankpins. These rubbing marks can be removed by going for 1.5 mm under size of crankpin.

11. After grinding, radius grinding, oil hole trimming and polishing carried in crankpin No 1,5 and 8.

ANNEXURE - I

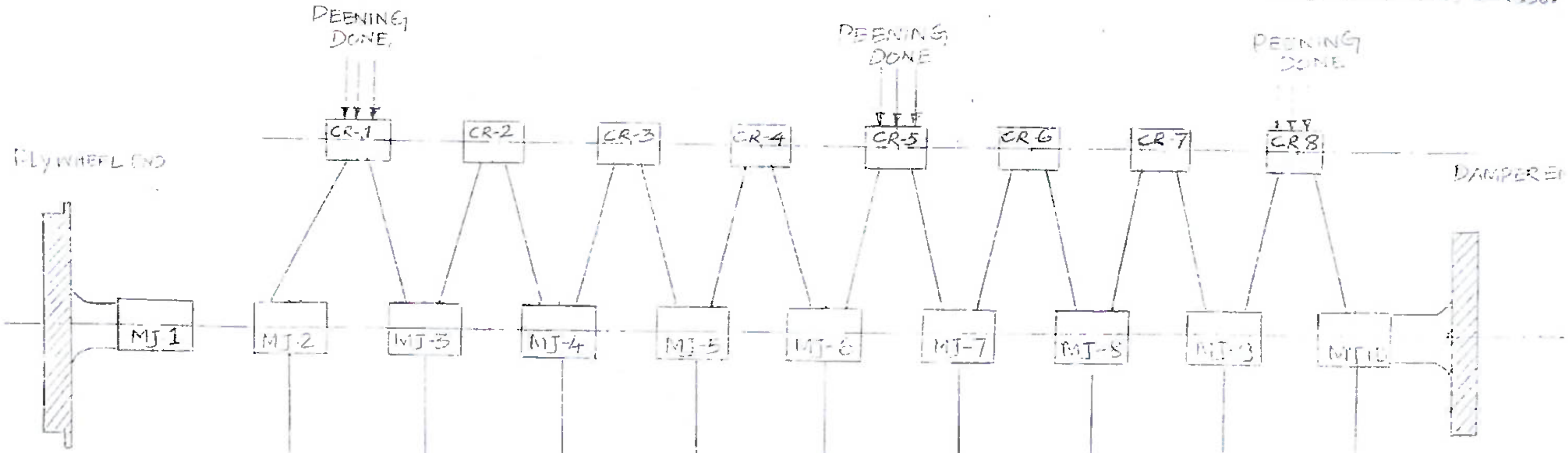
# EUROTECH SYSTEMS

## RUN OUT READING OF ALLEN ENGINE MP-II

ENGINE SL NO: DG/50101/[illegible]

CRANK SHAFT SL NO. MS&CO 3A6[illegible]

DBR 311316 [illegible]




| | MJ-2 | MJ-3 | MJ-4 | MJ-5 | MJ-6 | MJ-7 | MJ-8 | MJ-9 | MJ-10 |
|---|---|---|---|---|---|---|---|---|---|
| BEFORE PEENING → | 0.39mm | 0.47mm | 0.075mm | 0.13mm | 0.32mm | 0.31mm | 0.07mm | 0.05mm | 0.47mm |
| AFTER PEENING OF PIN NO 1,5&8 | 0.055mm | 0.065mm | 0.075mm | 0.02mm | 0.035mm | 0.05mm | 0.07mm | 0.075mm | 0.085mm |

STARTING REFERENCE : PIN NO 1 at TDC

DIRECTION OF ROTATION : CLOCKWISE WHEN VIEWING FROM DAMPER END

NOTE : After peening for clarification purpose the oil clearance of MJ-10 has been checked and found inbetween 0.21mm to 0.225mm. (ie within Accepeted limits)

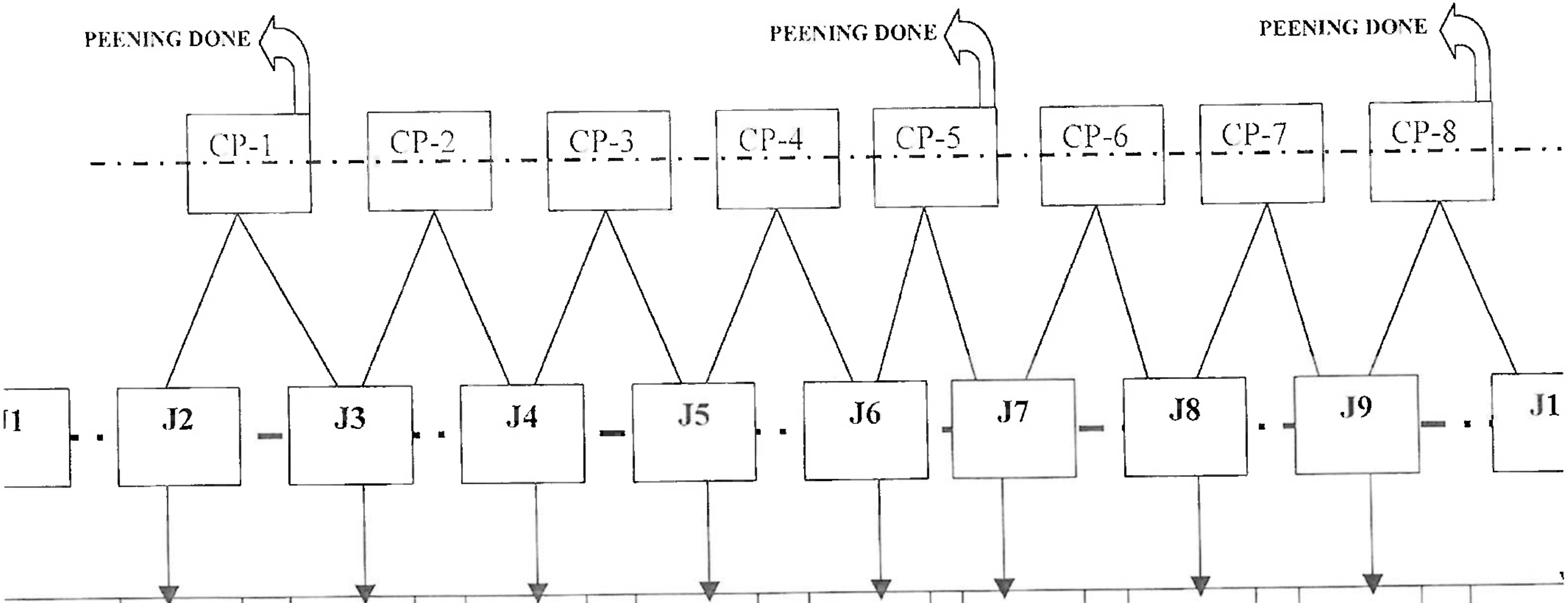
FOR EUROTECH SYSTEMS

FOR CUSTOMER

# RUN OUT READINGS OF MP-2 CRANK SHAFT, SMPL SIDDHPUR

(ENGINE SL. NO. D6/50161/1-1.

CRANK SHAFT SL NO. MS & CO 3A6201 LLOYD'S LIV DBR 311316 26089 SD 95507 ISS4)




| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| (in mm) | 0.39 | 0.47 | 0.075 | 0.13 | 0.32 | 0.31 | 0.07 | 0.08 | 0.4' |
| OF CP in mm) | 0.055 | 0.065 | 0.075 | 0.02 | 0.035 | 0.05 | 0.07 | 0.075 | 0.085 |

: 1) STARTING REFERENCE PIN NO. 1 AT TDC; 2) DIRECTION OF ROTATION: CLOCKWISE WHEN VIEWING FROM TDC; 3). M
ING CLEARANCE OF J 10 CHECKED AND FOUND 0.21 MM TO 0.225 MM.

**JAISWAL, SANJEEV (संजीव जायसवाल)** Sanjiv Kumar

| | |
|---|---|
| **From:** | JHA, S.K. (झा, एस. के.) |
| **Sent:** | Friday, July 03, 2015 2:24 PM |
| **To:** | BERA,T.K. (टी.के. बेरा) |
| **Cc:** | Suresh, P. (सुरेश, पी.); YADAV, B.D. (डा.बी.डी.यादव); Kumar, Ashok (कुमार, अशोक); JAISWAL, SANJEEV (संजीव जायसवाल); Sarkar, P. K. (सरकार, पी. के.); Tewari, A.K. (तिवारी, ए. के.); SINHA,JP (सिन्हा जे.पी.); Kumar, Tribhuwan (त्रिभुवन कुमार) |
| **Subject:** | RE: MP-2 SMPL Sidhpur - Breakdown Maintenance |
| **Attachments:** | Annexure-III (Preventive & Breakdown Maintenance History of MP-2).xlsx; Prelimin Report on MP-2 big end bearing failure.docx; Annexure-I (Free Spreadof big end brg).xlsx; Annexure-II (Scheduled Maintenance History of MP-2).xlsx |

Dear Sir,

After tripping of engine at High OMD% and subsequent observations, engine has been taken under 6000 hrs schedule maintenance which was also almost due. Prior to this breakdown, engine was being attended for welding of DE side seal cooling pipe and attending engine driven jacket water pump. Af attending these two issues, engine was re-started on 24.6.2015.

Investigation of the failure was carried out by a committee headed by CMNM, WRPL, Gauridad and committee visited Sidhpur on 1st and 2nd July 2015. Copy of the report along with annexure is enclos

Activities pertaining to 6000 hr maintenance is in progress. The newly procured lube oil filter elemer of radiator has also been delivered at site and same would be replaced along with ongoing maintenar An agency for inspecting/attending to the markings in crank pin no. 2 is also being lined up. MP-2 ha nos. 1mm undersized big end bearings and currently we have 2 bearings readily available as spare. A has been requested to expedite delivery of undersized big end bearing. Concurrently, already fitted 1 undersized bearing has been polished and made free of markings and is being kept ready.

Prevailing preventive/predictive maintenance practices in the station is also being checked for gaps, any and based on the findings suitable actions would be initiated. A quarterly review of mechanical maintenance with participation of all maintenance engineers is being started from this quarter to sha the practices and improvement required, if any.

Regards,

S.K.Jha
Viramgam

**From:** BERA,T.K. (टी.के. बेरा)
**Sent:** 30 June 2015 15:51
**To:** JHA, S.K. (झा, एस. के.)
**Cc:** Suresh, P. (सुरेश, पी.); YADAV, B.D. (डा.बी.डी.यादव); Kumar, Ashok (कुमार, अशोक); JAISWAL, SANJEEV (सं जायसवाल)
**Subject:** MP-2 SMPL Sidhpur - Breakdown Maintenance

MP2 at SMPL Sidhpur is under breakdown maintenance from 19.6.2015 initially breakdown was due to DE side leakage from seal cooling pipe. Though the problem was attended on 24.6.2015 and engine started. But the engine is again not available since 2307 hrs 24.6.2015 due to 100% deviation in cylinder no. 3

After that various activities is being carried out and w.e.f 29.6.2015 6000 hrs maintenance was taken. With this engine is not available for last 11 days.

In the daily reports activities details has been furnished but the cause of defect not indicated.

As discussed, you are requested to investigate the reasons of failure and furnish the report for management apprisal.

Regards

तापस कुमार बेरा / Tapas Kumar Bera
उप महाप्रबन्धक प्रभारी (अनुरक्षण) / DGM I/C (Maint)
इंडियन ऑयल कॉर्पोरेशन लिमिटेड/ Indian Oil Corporation Ltd.
पाइप.लाइन मुख्यालय., नॉएडा/ Pipeline Head Office, NOIDA
A-1 Udyog Marg, Sector-1, NOIDA
Tel Off (+91) 120 2448924, 2448884, Mob: 9582211342

# Breakdown analysis of MP-2 Engine at SMPL Sidhpur

## Introduction:

MP-2 has clocked a CRH of 93022 as on 24.06.2015.
MP-2 engine was running continuously since 14:52 Hrs on 24.06.2015.
The engine had tripped in OMD deviation 100% in Cylinder-3 at 23:07 Hrs on 24.06.2015.

**Parameters of the engine at 16:00 Hrs on 24.06.2015 (as taken from Log sheet) is:**
Lube oil pressure before filter-3.4 kg/cm2
Lube oil pressure after filter-3.1 kg/cm2
Lube oil filter diff press- 0.2 kg/cm2
Lube oil temp before cooler- 78 deg C
Lube oil temp after cooler- 55 deg C
OMD- 15% (in all cylinders)

**Parameters of the engine at 22:00 Hrs on 24.06.2015 (before tripping, as taken from log sheet) is:**
Lube oil pressure before filter-3.4 kg/cm2
Lube oil pressure after filter-3.0 kg/cm2
Lube oil filter diff press- 0.2 kg/cm2
Lube oil temp before cooler- 75 deg C
Lube oil temp after cooler- 55 deg C
OMD- 10% (in all cylinders)

## Observations after engine tripping in 100% OMD in Cyl-3:

1) After MP-2 tripped in 100% OMD in cyl-3 at 23:07 hrs, at first the temperature of big end bearing and main bearing of all cylinders were checked from crank case doors by temperature gun.
2) All crankcase doors were opened.
3) Temperature of Connecting rod-2 at big end bearing position and in the I-section was observed to be 95 deg C, but other big end bearing temperatures were found in the range of 78-82 deg C.
4) All Main bearing temperatures were noted and found to be in the range of 75-80 deg C range with no abnormalities.
5) Upon physical inspection, slight white metal of the big end bearing was observed over the crank pin which was stuck in the connecting rod lower half portion.

Since 6000 hrs scheduled maintenance of MP-2 at SMPL Sidhpur is due, it was decided to take the engine under 6000 hrs major overhaul including checking, inspection and corrective action for breakdown.

**Maintenance activity carried out after the Engine was taken under 6000 Hrs major overhaul as on 02.07.2015**

1) Cylinder head, piston-con rod assembly of cylinder-2 was removed.
2) Big end bearing-2 was removed and both the top and bottom bearing shells were found in its corresponding dowelling position of the con rod. Both the bearing shells did not rotate from its seating position in con rod.
3) Condition of cylinder head, liner and liner landing area of cylinder-2 upon physical inspection was found normal. No heating marks were observed.
4) Piston-con rod assembly of cylinder-2 was dismantled and piston plugs were found in place and 2 holes were open (in opposite end) which is indicative of the fact that there was no oil starvation in the piston.
5) Gudgeon pin-2 was removed and no abnormalities, heating marks, etc., were observed.
6) ***Checked big end bearing-2, observed 70-80% erosion and removal of white metal from the parent metal of the big end bearing.***
7) Connecting rod-2 was inspected and heating mark (10%) was observed on the top portion of big end (towards I-section)-flywheel side. Similar heating was also observed over the big end bearing shell (top half) of cylinder-2.
8) Crank pin-2 was inspected and white metal of the big end bearing was found stuck over the crank pin and scouring marks were observed. Cleaning of the same is under progress. Crank pin hole was checked and found normal.
9) End float in all the remaining connecting rods was checked and found normal.
10) Hardness mapping of cylinder-2 is done and a total of 54 readings were taken and the values obtained were within limits prescribed the engine manufacturer, i.e., less than 350 hB (in case of old/used crankshaft). However, since complete cleaning of crankpin-2 is still under progress, it was decided to again carry out hardness mapping of crankpin-2 and also of the remaining crankpins for comparison purpose after cleaning of crankpin-2 and dismantling of rest of the piston con rod assembly has been done.
11) To check for surface irregularities and cracks, it was also decided to carry out Dye penetration test crankpin-2, piston ring grooves and connecting rod big end bore of cylinder-2.
12) Furthermore, for the sake of completeness and ascertain integrity of the engine crankshaft, it was also decided to carry out Ultrasonic thickness check by Ultrasonic Flaw detector of all the crankpins.
13) Before complete cleaning of the crankpin-2 was done, the ovality of the crankpin (measured in both flywheel and damper side) was observed to be 0.04 mm, which is well within limits as prescribed by the engine manufacturer (0.102 mm). However, ovality of this crankpin will be rechecked again post cleaning.
14) To check for oil passage constriction, lube oil inlet piping of all the main bearings was removed and compressed air was blown into the inlet port of the main bearings and no blockage of air was observed in any of the crank pins. It may also be noted that the lube oil life of this engine is 3100 Hrs.
15) Also, as evident from site inspection; no abnormalities or any heating marks were observed in gudgeon pin, small end bush, push rod, crosshead, crosshead guide of the valve lever casing (exhaust side) which is the last point of lubrication in Allen supplied engines.
16) Condition of lube oil filter element was also inspected and no metal parts were found in it. The lube oil filter element (in both banks) was fully intact and found in position.

17) ***Marking in almost all other big end bearings is noticeable.***
18) Connecting rods of all the cylinders were removed and crankpins were cleaned and observed. No appreciable markings were found on other crankpins.
19) Since, appreciable quantity of white metal was found eroded from the big end bearing shell of cylinder-2; it was decided to ascertain the physical condition of the bearings and check for free spread of all the main bearings in this engine during the ongoing major overhaul before its reuse.
20) Free Spread measurement of all big end bearings were carried out and the measurement sheet is attached as Annexure-I. From the free spread values, it was observed that the free spread in Cylinder-2 and in Cylinder-8 was comparatively less and this was presumed due to removal of white metal in both the bearings (70-80% white metal removal in case of big end bearing-2 and partial white metal overlay was damaged in big end bearing-8) followed by temperature rise due to which the big end bearing has contracted due to sudden temperature drop after tripping of the engine. Although free spread in Big end bearing No:-1,3,4,5,6 & 7 were found within limits; it was decided to replace all the big end bearings of the engine since minor markings were observed in almost all the remaining bearings (as stated in point-17).
21) Hardness mapping of crankpin-2 and other crankpins was carried out and the results indicate maximum hardness value of 296 hB of crankpin-2.
22) Also, Hardness mapping of crankpin-8 was carried out and the values obtained are well within limits. For comparison purpose, hardness value checking of all the crankpins was done and the values obtained are in line with the manufacturer's tolerance limits.
23) Dye penetration test of crankpin-2 was carried out and no cracks/surface irregularities were observed in crankpin-2.
24) Dye penetration test of connecting rod-2 big end bore was also carried out and no cracks or surface porosities/undercuts, surface irregularities were visually observed.
25) Also, for the sake of completeness; dye penetration test was carried out of the top piston ring, pressure ring and oil scrapper ring grooves to ensure that there are no cracks in the ring grooves. No abnormalities or cracks were observed in any of the piston ring grooves.
26) Moreover, UT testing by UFD was also carried out in all the crankpins and while observing the spectra, no odd peaks were observed in the UFD, indicating that there are no cracks in the surface or in the bulk in any of the crankpins of MP-2 engine.
27) The crankpin-2 was again cleaned and ovality was re-measured in flywheel and damper position. Ovality value in Flywheel and Damper side of crankpin-2 is found to be 0.07 mm and 0.05 mm respectively, which is within the maximum allowable limit of 0.102 mm (as per M/s Allen Diesels).
28) The ovality in Connecting rod-2 was taken and the values obtained in Flywheel and Damper side are 0.03 mm and 0.04 mm respectively. Maximum taperness of the connecting rod is observed to be 0.02 mm. In this case also, the values obtained are well within permissible limits as prescribed by the engine manufacturer (0.10 mm for maximum ovality and 0.05 mm for maximum taperness of connecting rod).
29) The scheduled maintenance history of MP-2 (from 01.01.2013 till date) is attached as Annexure-II.
30) The preventive and breakdown maintenance history of MP-2 engine (from 01.12.2013 till date) is also attached as Annexure-III.

**Action plan:**

1. The high point and scouring marks on crank pin no.2 has to be removed by manual polishing. And if the desired result of surface smoothness is not achieved then we may deploy outside agency for honing.
2. The surface roughness $R_a$ value of crankpin-2 shall be checked and based on the result further course of action will be decided.
3. All the big end bearings will be replaced. Big end bearing 1, 5, & 8 are undersized bearings by 1mm. Only 2 nos. of new undersized bearings are available at Sidhpur. 3 nos. of undersized bearings of 8S37G Sr engines (1 mm) are already under procurement which may take 1 and half months.
4. Condition of all the main bearings to be ascertained after dismantling.
5. Although the heating mark in the big end bore of connecting rod-2 (towards I-section) was removed after polishing and cleaning with emery paper and both ovality and taperness of the connecting rod-2 are within limits, it is advised to replace the connecting rod of cylinder-2 with a reconditioned one and send the dismantled con rod to Central Maintenance, Gauridad for further checking of straightness and analysis.
6. The hardness mapping of crank pin -2 has been done (a total of 192 readings were taken). The highest value was observed to be 296 hB and minimum readings was 215 hB. The readings are well within the acceptable limit of 350 hB (for old/used crankshaft). The maximum hardness (296 hB) was observed at a point which is at an angle of 44.5 degree clockwise from the crankpin hole viewed from flywheel end.
7. The white metal in big end bearing of cylinder no. 8 was also found slightly eroded off. Reason for the same may be attributed to the eroded off white metal from big end bearing-2 which has damaged the rest of the bearings.

**Probable causes of failure:**

1. No apparent causes of failure have been observed. The lube oil passage was checked by blowing air and found normal. No lube oil flow restriction was observed.
2. Due to white metal erosion in big end bearing-2, minor wear and markings were observed in all the other bearings.
3. During the last 12000 hrs scheduled maintenance of the engine, all the connecting rods were checked for taperness, ovality and minimum bore criteria. All the readings were well within limits prescribed by the engine manufacturer. However, there may be a possibility of twisting of the connecting rod during its operation and the same will be sent to Rajkot for further checking and analysis.

| | | |
|---|---|---|
| **(P.K. Sarkar)** | **(Tribhuwan Kumar)** | **(Parichay Das)** |
| CMNM, WRPL Rajkot | SOM, WRPL, Viramgam | SO&ME, WRPL, Sidhpur |

Annexure-I

**Free Spread Measurement Sheet of MP-2 Big end bearing (Post failure)**

| Cylinder No | Big end bearing | Position | Value (mm) | Condition | Remarks |
|---|---|---|---|---|---|
| 1 | 1 | Top | 252.10 | Minor markings were observed | To be replaced with new one (1 mm undersized bearing to be used since this crankpin was already machined to NB 239.00 mm). |
| | | Bottom | 252.10 | | |
| 2 | 2 | Top | 242.00 | White metal eroded off in almost 70-80% of the bearing area | To be replaced with new one. |
| | | Bottom | 242.00 | | |
| 3 | 3 | Top | 255.50 | Minor markings were observed. | To be replaced with new one. |
| | | Bottom | 255.50 | | |
| 4 | 4 | Top | 254.00 | Very less markings were observed. | To be replaced with new one. |
| | | Bottom | 254.00 | | |
| 5 | 5 | Top | 254.25 | Minor markings were observed. | To be replaced with new one (1 mm undersized bearing to be used since this crankpin was already machined to NB 239.00 mm). |
| | | Bottom | 254.25 | | |
| 6 | 6 | Top | 253.80 | Very less markings were observed. | To be replaced with new one. |
| | | Bottom | 253.90 | | |
| 7 | 7 | Top | 254.00 | Very less markings were observed. | To be replaced with new one. |
| | | Bottom | 254.20 | | |
| 8 | 8 | Top | 247.00 | White metal overlay has been found damaged. | To be replaced with new one (1 mm undersized bearing to be used since this crankpin was already machined to NB 239.00 mm). |
| | | Bottom | 247.15 | | |